SARATH AC

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ABOUT ME

Enthusiastic Data Scientist with a solid foundation in data analytics and visualization. Skilled in data cleaning, EDA, and generating insights to support decisions. Proficient in Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, and Excel, with experience in creating dashboards in Power BI/Tableau. Passionate about turning complex data into clear, useful insights.

EDUCATION

B-Tech | Computer Science and Engineering

College Of Engineering Chengannur| 2020-2024

- 12 week Advanced python computing course, IIT Madras (NPTEL)

Upskill Course| Full Stack Data Science

AlmaBetter| 2024 November -2025 August

WORK EXPERIENCE

INTERNSHIP

Centre for Development of Advanced Computing (CDAC) | 2025 Aug – 2026 Jan Trivandrum

- Contributed to a **Brain–Machine Interface (BMI)** web application for end-to-end EEG data ingestion, processing, visualization, and analysis.
- Designed and implemented robust **EEG data upload pipelines**, supporting **EDF format ingestion** and **conversion to DICOM** for standardized medical data handling.
- Developed a **custom EEG Viewer**, functionally comparable to the **Natus EEG application**, seamlessly integrated into the web platform.
- Implemented advanced EEG viewer controls, including **channel selection**, configurable **sensitivity, spacing, time base, and time-per-window**, and comprehensive **signal filtering** (notch, low-pass, high-pass, band-pass, bandstop).
- Built signal processing and analytical features such as **FFT**, **Power Spectral Density (PSD)**, and **montage configuration** for in-depth EEG interpretation.
- Enabled **offline streaming and playback** of selected EEG segments for retrospective analysis.
- Implemented **real-time EEG data acquisition and visualization** by streaming signals from EEG hardware using **WebSocket-based communication**.
- Developed interactive EEG analysis dashboards using **heatmaps, bar charts, and line charts** to visualize signal density, data quality, and event frequency across datasets.
- Worked as a **full-stack developer** using **React.js** for frontend interfaces and **Python (Django)** for backend services, integrating **REST APIs**, **JavaScript**, and **CSS** for scalable and responsive system design.

PROJECTS

End-to-End ML Project: Student Exam Score Prediction

- Developed an end-to-end machine learning pipeline in Python to predict student exam scores, covering data ingestion, preprocessing, model training, evaluation, and inference.
- Implemented data ingestion and feature preprocessing using Scikit-learn's ColumnTransformer to handle mixed data types and ensure reproducible train–test splits.
- Trained and optimized XGBoost and CatBoost regression models , achieving a best test R^2 score of ~0.87.
- Conducted systematic multi-model evaluation using train/test R^2 metrics to compare ensemble and boosting regressors and select the optimal model.
- Designed a modular, production-ready architecture with robust logging and exception handling, enabling traceable workflows and reliable inference.

End-to-End Bank Loan Eligibility Analysis

- Analyzed financial data of 3,000 customers to identify factors linked to loan default. Performed data cleaning (handling missing values, removing duplicates, encoding categorical variables, outlier detection) using Pandas & NumPy.
- Conducted feature engineering (binning income levels, debt-to-income ratio, credit score grouping) to improve prediction accuracy.
- Performed Exploratory Data Analysis (EDA) with Matplotlib & Seaborn to visualize patterns in approval rates, defaults, and customer profiles.
- Designed an interactive Power BI dashboard for stakeholders to monitor approval trends, high-risk segments, and default likelihood.

SKILLS

programming languages : Python , C programming , React.js

Database : MySQL,

Frameworks : Djano,RestApi

Machine learning algorithms : Linear regression, Logistic, regression, Decision tree, Random forest, XGBoost

Libraries : Pandas,NumPy,Matplotlib,Seaborn

Tools/Technologies : AWS,Git

Core concepts : Data Structures and algorithm, Mechine Learning , Deep Learning, Exploratory Data Analysis,Data Mining, Data Cleaning

CERTIFFICATES

Google Data Analyst professional certificate (Coursera)

Full Stack Data Science